Migration and Transfer

Moving data into AWS

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DataCumulus Sundog Education

AWS Application Discovery Service

- Plan migration projects by gathering information about on-premises data centers
- Server utilization data and dependency mapping are important for migrations

- **Agentless Discovery (AWS Agentless Discovery Connector)**
  - VM inventory, configuration, and performance history such as CPU, memory, and disk usage
- **Agent-based Discovery (AWS Application Discovery Agent)**
  - System configuration, system performance, running processes, and details of the network connections between systems

- Resulting data can be viewed within AWS Migration Hub

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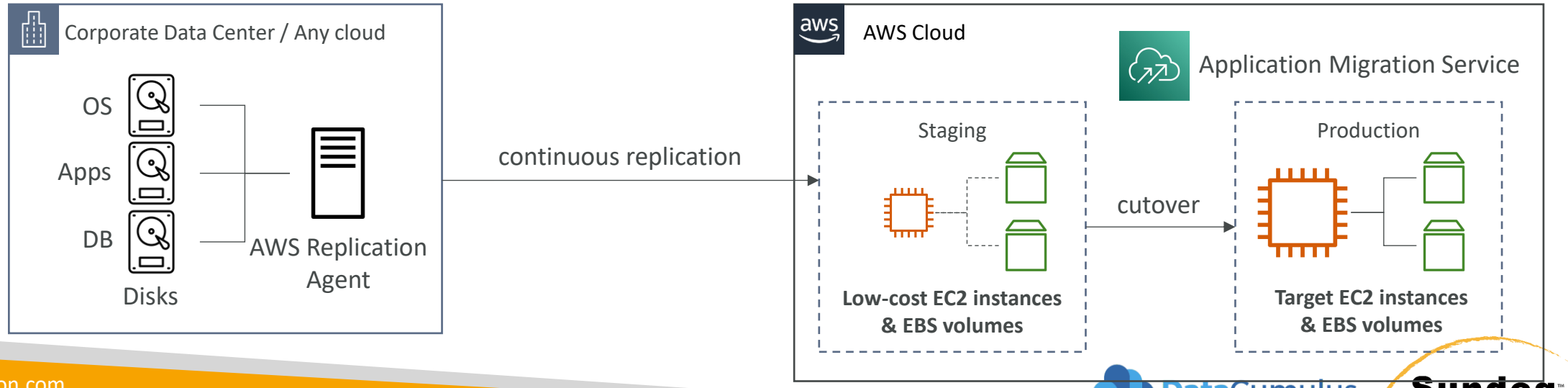
AWS Application Migration Service (MGN)

- *The "AWS evolution" of CloudEndure Migration, replacing AWS Server Migration Service (SMS)*

- Lift-and-shift (rehost) solution which simplify **migrating** applications to AWS
- Converts your physical, virtual, and cloud-based servers to run natively on AWS
- Supports wide range of platforms, Operating Systems, and databases
- Minimal downtime, reduced costs



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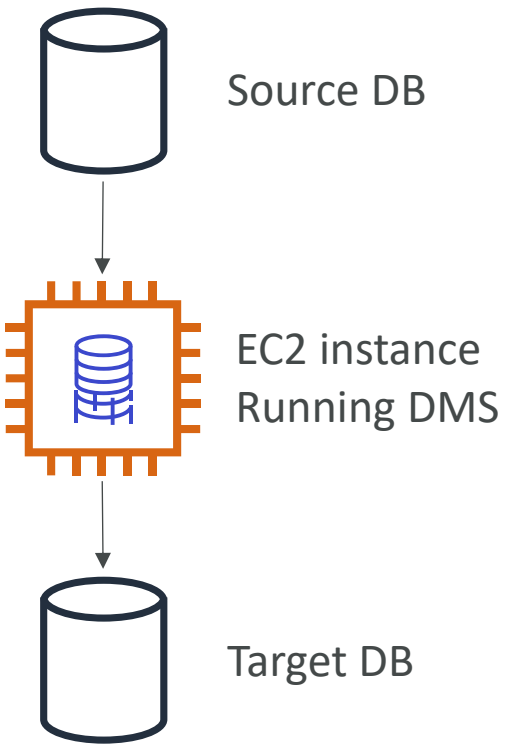
DMS – Database Migration Service

- Quickly and securely migrate databases to AWS, resilient, self healing
- The source database remains available during the migration
- Supports:
  - Homogeneous migrations: ex Oracle to Oracle
  - Heterogeneous migrations: ex Microsoft SQL Server to Aurora
- Continuous Data Replication using CDC
- You must create an EC2 instance to perform the replication tasks



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DMS Sources and Targets

SOURCES:

- On-Premises and EC2 instances databases: *Oracle, MS SQL Server, MySQL, MariaDB, PostgreSQL, MongoDB, SAP, DB2*
- Azure: *Azure SQL Database*
- Amazon RDS: all including Aurora
- Amazon S3
- DocumentDB

TARGETS:

- On-Premises and EC2 instances databases: Oracle, MS SQL Server, MySQL, MariaDB, PostgreSQL, SAP
- Amazon RDS
- Redshift, DynamoDB, S3
- OpenSearch Service
- Kinesis Data Streams
- Apache Kafka
- DocumentDB & Amazon Neptune
- Redis & Babelfish

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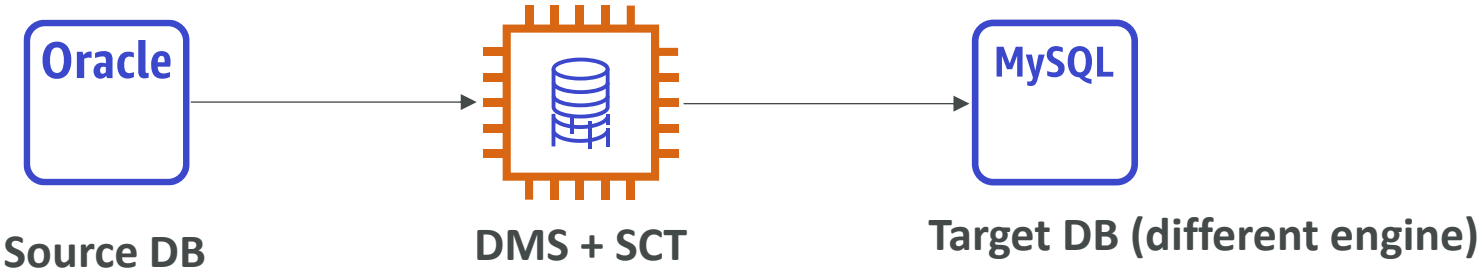
AWS Schema Conversion Tool (SCT)

- Convert your Database's Schema from one engine to another
- Example OLTP: (SQL Server or Oracle) to MySQL, PostgreSQL, Aurora
- Example OLAP: (Teradata or Oracle) to Amazon Redshift
- Prefer compute-intensive instances to optimize data conversions



- **You do not need to use SCT if you are migrating the same DB engine**
  - Ex: On-Premise PostgreSQL => RDS PostgreSQL
  - The DB engine is still PostgreSQL (RDS is the platform)

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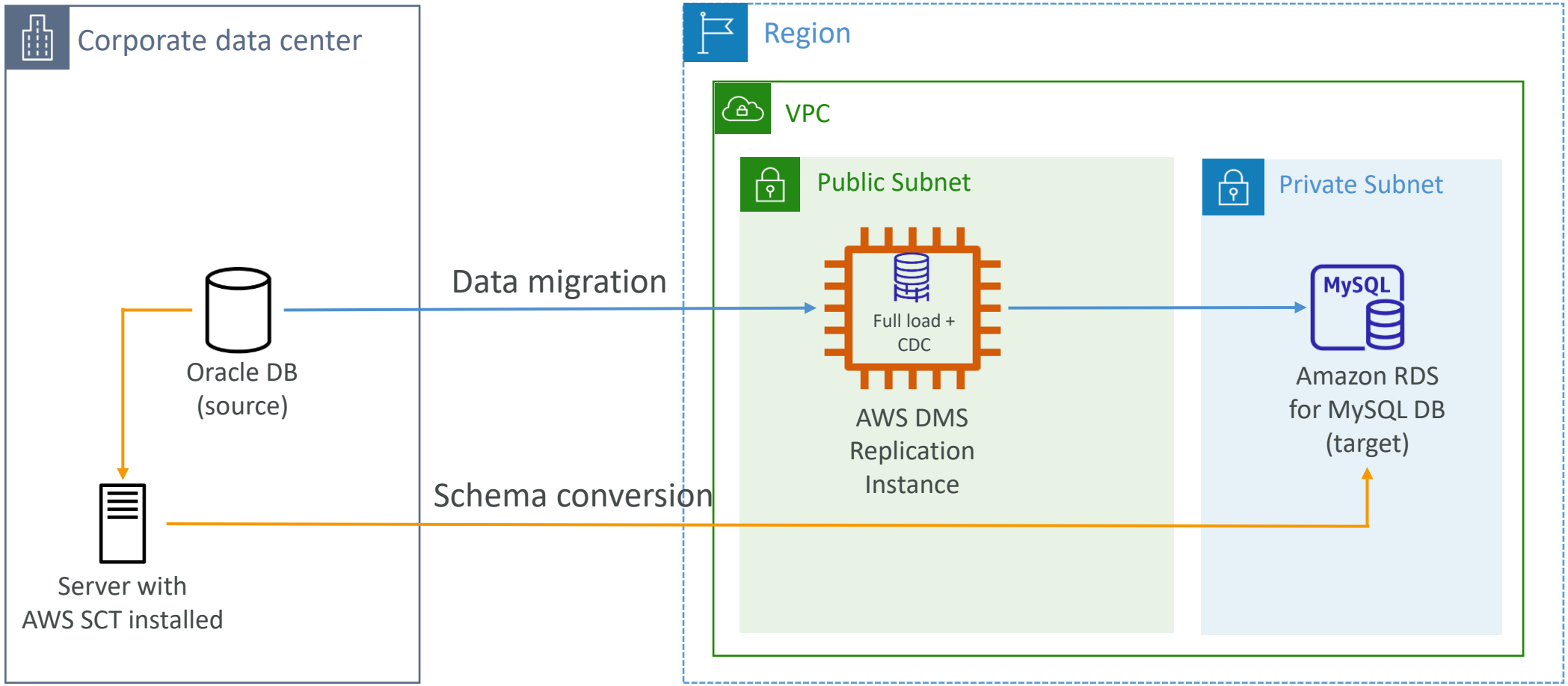
DMS - Continuous Replication



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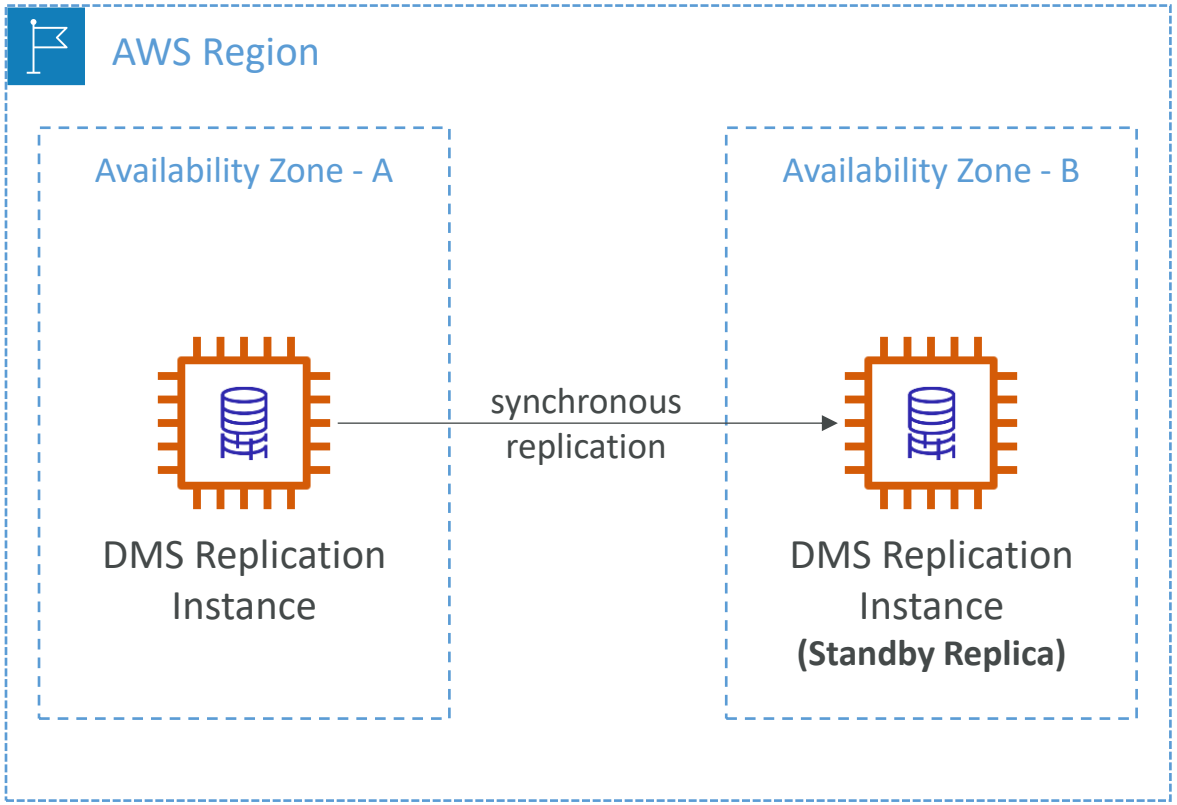
AWS DMS – Multi-AZ Deployment

- When Multi-AZ Enabled, DMS provisions and maintains a synchronously stand replica in a different AZ

- Advantages:
  - Provides Data Redundancy
  - Eliminates I/O freezes
  - Minimizes latency spikes



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AWS DataSync

- Move large amount of data to and from
  - On-premises / other cloud to AWS (NFS, SMB, HDFS, S3 API…) – **needs agent**
  - AWS to AWS (different storage services) – no agent needed
- Can synchronize to:
  - Amazon S3 (any storage classes – including Glacier)
  - Amazon EFS
  - Amazon FSx (Windows, Lustre, NetApp, OpenZFS...)
- Replication tasks can be scheduled hourly, daily, weekly
- **File permissions and metadata are preserved** (NFS POSIX, SMB…)
- One agent task can use 10 Gbps, can setup a bandwidth limit

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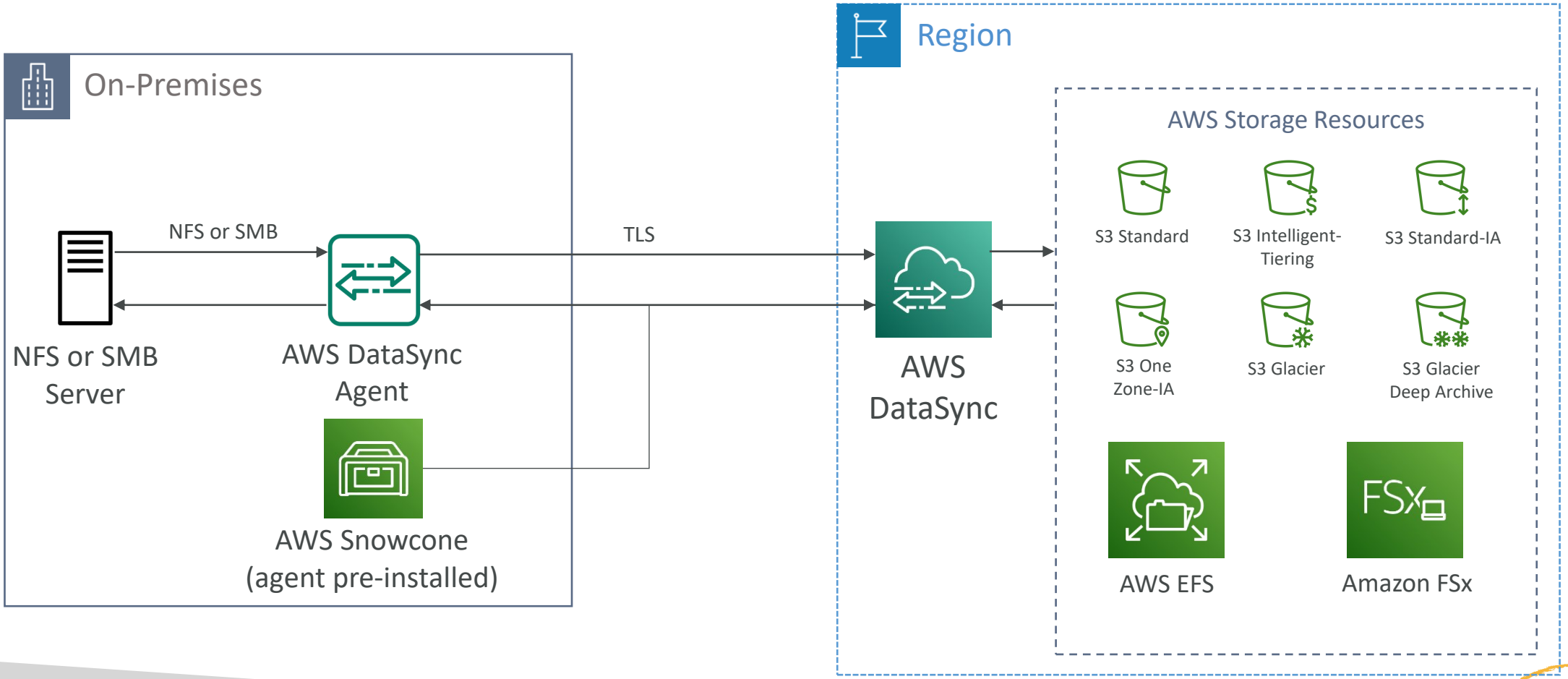
AWS DataSync
NFS / SMB to AWS (S3, EFS, FSx…)



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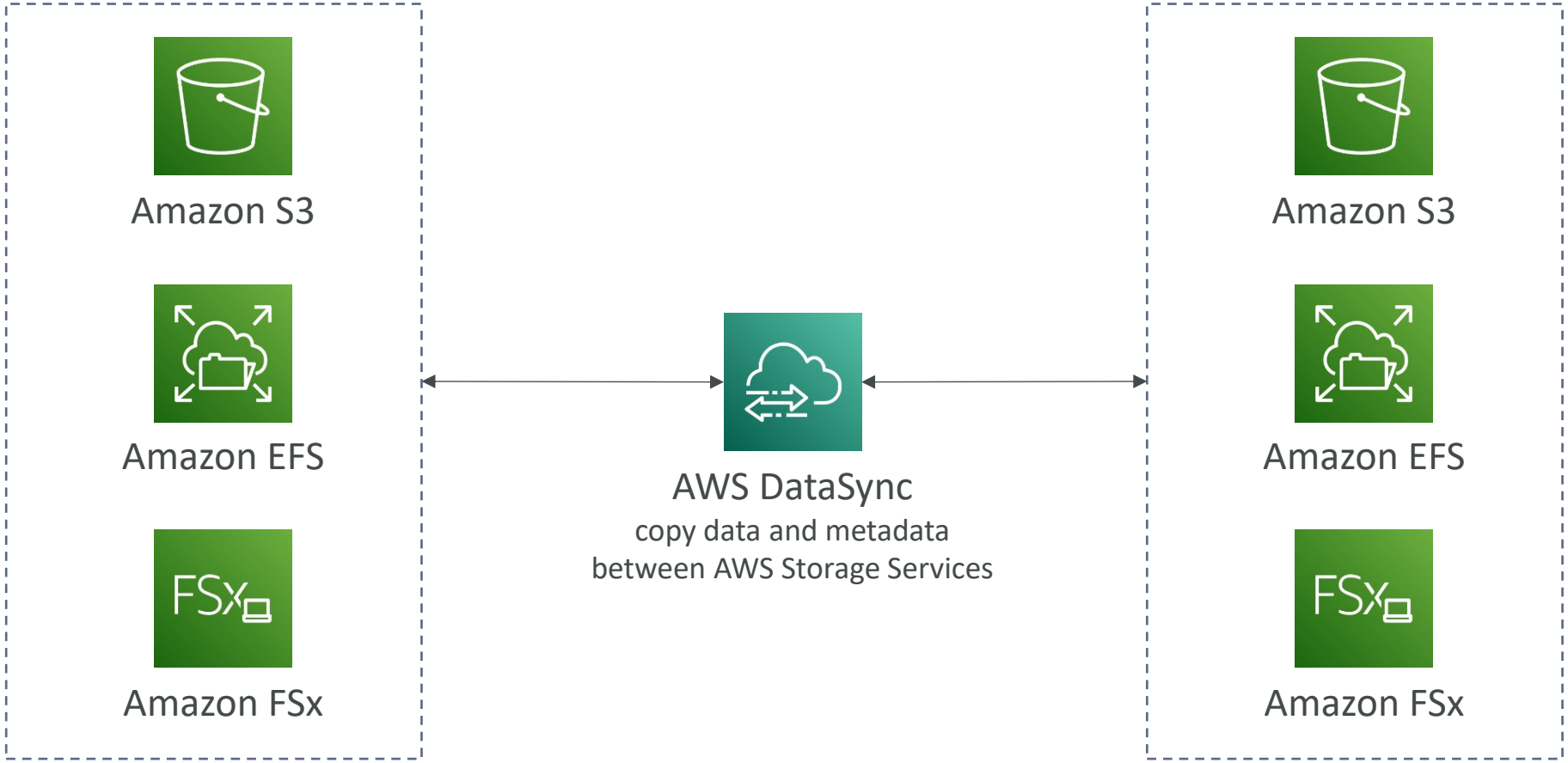
AWS DataSync
Transfer between AWS storage services



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AWS Snowball

- Highly-secure, portable devices to **collect and process data at the edge**, and **migrate data into and out of AWS**
- Helps migrate up to Petabytes of data

Device	Compute	Memory	Storage (SSD)
Snowball Edge **Storage** Optimized	104 vCPUs	416 GB	**210 TB**
Snowball Edge **Compute** Optimized	104 vCPUs	416 GB	**28 TB**

Snowball Edge

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Data Migrations with Snowball

	Time to Transfer		
	100 Mbps	**1Gbps**	**10Gbps**
10 TB	12 days	30 hours	3 hours
100 TB	124 days	12 days	30 hours
1 PB	3 years	124 days	12 days

Challenges:

- Limited connectivity
- Limited bandwidth
- High network cost
- Shared bandwidth (can't maximize the line)
- Connection stability

AWS Snowball: offline devices to perform data migrations

If it takes more than a week to transfer over the network, use Snowball devices!

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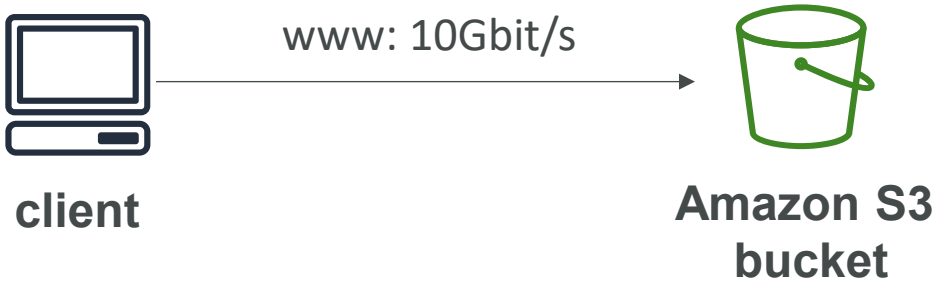
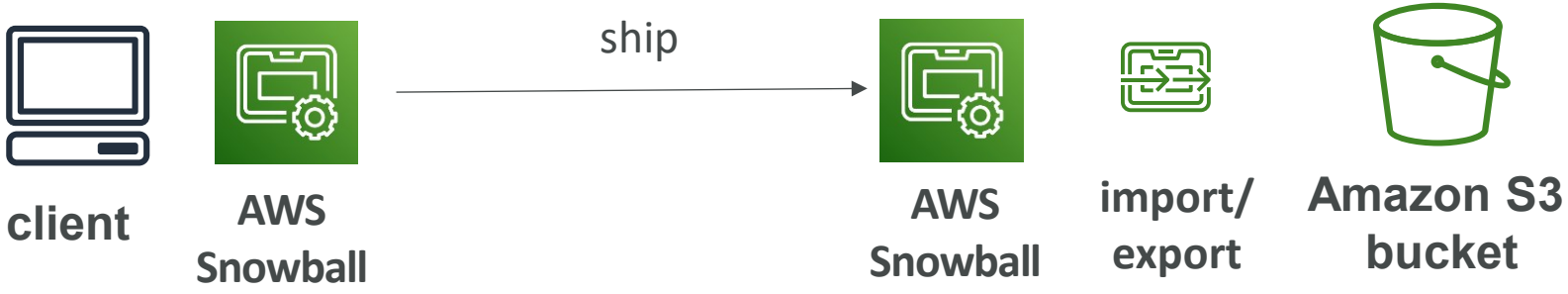
Diagrams

- Direct upload to S3:



- With Snowball:



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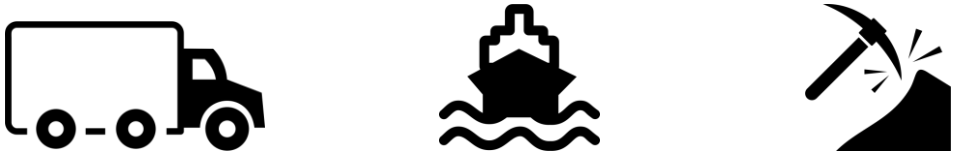
What is Edge Computing?

- Process data while it's being created on **an edge location**
  - A truck on the road, a ship on the sea, a mining station underground...



- These locations may have limited internet and no access to computing power
- We setup a **Snowball Edge** device to do edge computing
  - Snowball Edge Compute Optimized (dedicated for that use case) & Storage Optimized
  - Run EC2 Instances or Lambda functions at the edge
- Use cases: preprocess data, machine learning, transcoding media

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AWS Transfer Family

- A fully-managed service for file transfers into and out of Amazon S3 or Amazon EFS using the FTP protocol
- Supported Protocols
  - **AWS Transfer for FTP** (File Transfer Protocol (FTP))
  - **AWS Transfer for FTPS** (File Transfer Protocol over SSL (FTPS))
  - **AWS Transfer for SFTP** (Secure File Transfer Protocol (SFTP))
- Managed infrastructure, Scalable, Reliable, Highly Available (multi-AZ)
- Pay per provisioned endpoint per hour + data transfers in GB
- Store and manage users' credentials within the service
- Integrate with existing authentication systems (Microsoft Active Directory, LDAP, Okta, Amazon Cognito, custom)
- Usage: sharing files, public datasets, CRM, ERP, …

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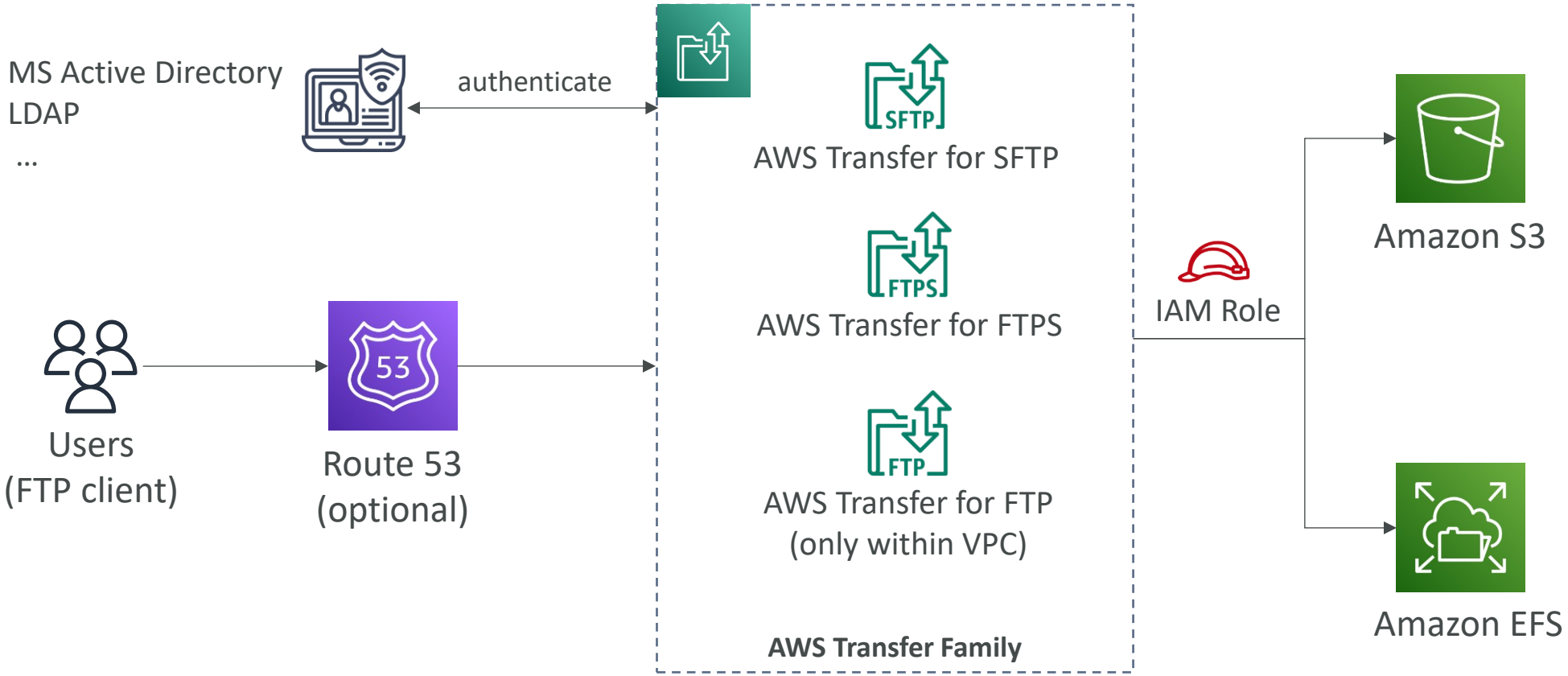
AWS Transfer Family



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